BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Binary Search

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. Does linear search require a sorted list?

2. Which is a physical security measure?

A) A strong password
B) File compression
C) A locked server room
D) A software licence

3. "The Data Protection Act permits unlimited retention and sharing of all pupil data." Correct this. Explain why.

4. A pupil opens another pupil's account without permission but changes nothing. "Access is legal if no files are deleted." Correct this. Why could it still break the law?

Today's objectives

State that binary search requires sorted data. Trace low, high and mid values correctly.

Key vocabulary - Read, cover, recall

Binary search
Search method repeatedly checking the midpoint of a sorted range.

Sorted
Arranged in order, such as ascending values.

Low index
Lowest index still being searched.

High index
Highest index still being searched.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

Sorted prerequisite	Binary search relies on sorted data. Without sorting, discarding half the list may incorrectly discard the target.
Find the midpoint	Keep low and high indices of the remaining range; calculate mid = (low + high) DIV 2 using integer division.
Discard half appropriately	If target is greater than the midpoint value, set low = mid + 1. If smaller, set high = mid - 1.
Repeat until found or empty	Compare the midpoint again until target is found, or low becomes greater than high, indicating no match.

I Do - One worked example

A sorted array is [2,4,6,8,10,12,14]. Trace binary search for 12 using low, high and mid. [4 marks]

1. Sorted prerequisite is satisfied; low = 0 and high = 6 [1].

2. mid = (0+6) DIV 2 = 3, value 8 [1].

3. 12 > 8, so discard indexes 0-3; now low = 4, high = 6 [1].

4. mid = (4+6) DIV 2 = 5, value 12, so return index 5 [1].

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Trace binary search for 9 in sorted [1,3,5,7,9], noting each midpoint. [4 marks]

2. Explain what happens if binary search is used on [10,3,8,1] without sorting. [2 marks]

3. Trace search for target 6 in sorted [1,2,4,7,9] and explain the unsuccessful result. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the assignment so the search moves right after comparing the middle value.

```
array weights = [1,2,4,8,16]
target = 16
low = 0
high = 4
mid = (low + high) DIV 2
if weights[mid] < target then
...
endif
print(low)
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. State the required property of an array before binary search can be used correctly. [1 mark]

Q3. For sorted [2,4,8,10,14], trace binary search for 14, showing each midpoint and discarded half. [4 marks]

Q4. Explain how binary search finds 5 in sorted [1,3,5,7,9,11,13], why sorting is essential and what happens when a target is missing. [6 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Binary search works reliably on any unsorted list. ☐ T ☐ F

2. Binary search relies on sorted data ☐ T ☐ F

3. Keep low and high indices of the remaining range; calculate mid = (low + high) DIV 2 using integer divis... ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 No; it examines items in sequence until a match is found or the list ends. ☐

Q2 C - A locked server room ☐

Q3 It sets rules for using personal data lawfully and protects people's rights. ☐

Q4 Access without permission can break the Computer Misuse Act, even if no files change. A law or licence sets what is allowed; having a copy does not give unlimited rights. ☐

Code First - Model

```
array weights = [1,2,4,8,16]
target = 16
low = 0
high = 4
mid = (low + high) DIV 2
if weights[mid] < target then
low = mid + 1
endif
print(low)
```

Output: 3

We Do - Check the model

1. low 0, high 4 gives mid 2, value 5 [1]. Since 9 > 5, low = 3 [1]. mid (3+4) DIV 2 = 3 gives 7 [1]; low = 4 and index 4 contains 9 [1]. ☐

2. The list is not sorted [1], so discarding half based on a midpoint comparison may wrongly exclude the target [1]. ☐

3. First mid index 2 gives value 4 [1], so low becomes 3 [1]. Next mid index 3 gives 7 [1], so high becomes 2 and low > high; target is absent [1]. ☐

Exit Ticket - Check

1. FALSE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Values must be in sorted order [1]. ☐

Q3 Mid = 2, value 8 [1]; target 14 > 8 so low = 3 [1]; next mid = 3, value 10 [1]; target 14 >10 so index 4 is checked and found [1]. ☐

Q4 Begin with low 0 high 6 [1], midpoint index 3 value 7 [1]. Target 5 is smaller so reduce high to 2 [1]. New midpoint index 1 value 3, so move low to 2 [1]; index 2 contains 5 [1]. Sorted data is essential because only then can half be discarded safely [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: